RAZAQ KHAN MOHAMMAD ABDUL

mohammadabdul.r@northeastern.edu | +1(857)395-4409 | linkedin.com/in/razaqkhanma | github.com/Khan-razaq | Boston, MA

SKILLS

Programming: C, C++, Python, Embedded C, MATLAB, Verilog

Frameworks & Algorithms: ROS, ROS2, Gazebo, MoveIt, PCL, OpenCV, ORB-SLAM, Navigation Stack, gmapping, AMCL, YOLO, tf2, TensorFlow, PyTorch, Sensor Fusion, Path Planning, PID, Pure Pursuit

Tools: Linux, Jupyter/Voila, Docker, Git, GitHub, CMake, Jira, SQL

Hardware: Jetson, RealSense D455, LIDAR, RTK-GPS, Raspberry Pi, Arduino, ATMega, PLCs, KiCAD, CAN, UART, SPI, I2C, RS485

WORK EXPERIENCE

Robotics Field Engineer, *Directed Machines, Reynolds, IN* **March 2026 – July 2026**

- Build waypoint navigation plans for AMRs across a 2,500-acre solar farm, tuning panel-post clearance and obstacle-detection width so the fleet mows between panel rows without collisions
- Operate a 7-robot fleet, driving 15 km of autonomous mowing per day and completing a 13-acre inverter block per week
- Diagnose and resolve sensor calibration and connectivity faults remotely, perform mechanical and electrical repairs across drivetrain, mower, and antenna systems, and track issues through GitHub to keep all 7 robots operational

Robotics Engineer Co-op, *Clarapath, Hawthorne, NY* **May 2024 - December 2024**

- Automated firmware deployment for Jetson devices by building a Python tool with server integration, cutting it from 8 hrs to 30 mins and enabling non-technical assemblers to flash units without Linux commands
- Built diagnostic and production-validation tools for the multi-motor subsystem, developing Voila GUIs with multithreading, SQL, and hardware control to command motors, read encoders, and verify 30+ checkpoints, cutting per-machine validation from 4 hrs to 10 mins
- Integrated PLC-controlled door sensors and latches into pathology equipment by modifying C++ control algorithms with ROS-based fail-safe logic, preventing operator exposure to blades and UV light

Robotics Engineer*, Dhi Sathi Robotics Pvt. Ltd. (Farm Sathi), India* **July 2022 - June 2023**

- Developed a ROS-based RTK-GPS waypoint navigation system for an agricultural robot, recording u-blox waypoints and prototyping a Pixhawk/ArduPilot controller to advance it from teleoperation toward autonomy
- Built a modular RS485 network connecting three subsystems (tiller, sprayer, and plower), writing embedded firmware for inter-module comms and fabricating production-ready electrical boxes

Robotics Intern*, Introbotics Systems Pvt. Ltd., India* **January 2022 - July 2022**

- Developed an autonomous navigation system for an indoor logistics AMR, integrating RPLidar with ROS navigation stack using gmapping for SLAM and AMCL for localization, achieving reliable obstacle avoidance
- Built a vision-guided manipulation system to pick and place target objects on a tabletop, using OpenCV/YOLO for object detection and MoveIt to control a 5-DOF arm

ACADEMIC PROJECTS

Terrain Perception for Quadruped Robots*, NEU* **February 2025 - April 2025**

- Benchmarked a ROS2/RealSense elevation-mapping pipeline across Jetson Nano and a laptop, using Docker and multithreading, profiling frame rate, CPU, and thermals to assess real-time feasibility on edge hardware
- Built a Gazebo quadruped simulation with a custom URDF to generate elevation maps toward CNN-based foothold detection for walkable-terrain assessment

Computer Vision and Deep Learning for Image Classification*, NEU* **January 2024 - April 2024**

- Built an OpenCV pipeline with filters, histogram-based retrieval, and real-time segmentation for face detection
- Trained ML models for visual recognition: CNN with transfer learning for digit-to-Greek letter classification and VGG16 waste classifier achieving 94.5% accuracy for real-time biodegradable detection via webcam

Autonomous Search-and-Rescue Robot*, NEU* **October 2023 - December 2023**

- Built a ROS SLAM and autonomous navigation system to map and explore an indoor environment for search and rescue
- Integrated AprilTag detection to identify and localize victim markers across the map, flagging positions for recovery

EDUCATION

Master of Science in Robotics, *Northeastern University, Boston, MA* **September 2023 - December 2025**

Bachelor of Engineering in Electronics and Communication, *Osmania University, India* **August 2018 - June 2022**

ACHIEVEMENTS

Published "ROS Based Autonomous Mobile Manipulator Robot" - Atlantis Press, Springer Nature; won first prize in the "Dr. Abdul Kalam Innovation Challenge"